stellar structure in terms of formulas instead of just numbers off a calculator. A single formula can embody all the information contained in many many tables of numbers. If he could get the right formula, it would contain simultaneously the structures of stars of 1 solar mass, 2 solar masses, 5 solar masses—any mass at all. But even with his brilliant mathematical skills, Tolman was unable to solve Volkoff's equation in terms of formulas.

"On the other hand," Tolman presumably argued to himself, "we know that the correct equation of state is not really the one Volkoff is using. Volkoff has ignored the nuclear force; and since we don't know the details of that force at high densities, we don't know the correct equation of state. So let me ask a different question from Volkoff. Let me ask how the masses of neutron stars depend on the equation of state. Let me pretend that the equation of state is very 'stiff,' that is, that it gives exceptionally high pressures, and let me ask what the neutron-star masses would be in that case. And then let me pretend the equation of state is very 'soft,' that is, that it gives exceptionally low pressures, and ask what then would be the neutron-star masses. In each case, I will adjust the hypothetical equation of state into a form for which I can solve Volkoff's differential equation in formulas. Though the equation of state I use will almost certainly not be the right one, my calculation will still give me a general idea of what the neutron-star masses might be if nature happens to choose a stiff equation of state, and what they might be if nature chooses a soft one."

On 19 October, Tolman sent a long letter to Oppenheimer describing some of the stellar-structure formulas and neutron-star masses he had derived for several hypothetical equations of state. A week or so later, Oppenheimer drove down to Pasadena to spend a few days talking with Tolman about the project. On 9 November, Tolman wrote Oppenheimer another long letter, with more formulas. In the meantime, Volkoff was punching away on his Marchant buttons. In early December, Volkoff finished. He had numerical models for neutron stars with masses 0.3, 0.6, and 0.7 solar mass; and he had found that, *if there were no nuclear force in our Universe, then neutron stars would always be less massive than 0.7 solar mass.*

What a surprise! Oppenheimer's crude estimate, before Volkoff started computing, had been a maximum mass of 6 solar masses. To protect massive stars against forming black holes, the careful calculation would